Mid Semester Examination – Third semester-2025
Center for Health Science and Technology (JISIASR Kolkata)

Program Name: M.Sc. in Medical Biotechnology and Bioinformatics
Semester: Second
Paper Name: Database Design
Paper Code: PDS2007
Date (Time): 13.03.2026 (1pm - 2.30pm)
Full Marks: 30

Attempt all questions

Group A (5x2=10)

1. Define super key.
2. What is the difference between a primary key and a super key?
3. What is weak entity set and how is it represented in ER diagram
4. In the ER diagram, how the manager entity set should be connected to the employee entity set.
5. What is a one-to-one and many-to-one relationship? Give an example of each.

Group B (2x5=10)

6. Consider the employee database of Figure 2.17. Give an expression in the relational algebra to express each of the following queries. a. Find the name of each employee who lives in the city "Miami". b. Find the name of each employee whose salary is greater than $100000. c. Find the name of each employee who lives in "Miami" and whose salary is greater than $100,000.

7. a. In the instance of the instructor shown in Figure 2.1, no two instructors have the same name. From this, can we conclude that name can be used as a superkey (or primary key) of an instructor? b. Consider the time slot relation. Given that a particular time slot can meet more than once in a week, explain why day and start time are part of the primary key of this relation, while end time is not.

Group C (1x10=10)

8. What is the result of first performing the Cartesian product of student and advisor, and then performing a selection operation on the result with the predicate s_id = ID? (Using the symbolic notation of relational algebra, this query can be written as $\sigma_{s_id=ID}(student \times advisor)$.) Explain with example.

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employee (*person_name*, *street*, *city*)
works (*person_name*, *company_name*, *salary*)
company (*company_name*, *city*)

Figure 2.17 Employee database.

ID	*name*	*dept_name*	*salary*
10101	Srinivasan	Comp. Sci.	65000
12121	Wu	Finance	90000
15151	Mozart	Music	40000
22222	Einstein	Physics	95000
32343	El Said	History	60000
33456	Gold	Physics	87000
45565	Katz	Comp. Sci.	75000
58583	Califieri	History	62000
76543	Singh	Finance	80000
76766	Crick	Biology	72000
83821	Brandt	Comp. Sci.	92000
98345	Kim	Elec. Eng.	80000

Figure 2.1 The *instructor* relation.

classroom(*building*, *room_number*, *capacity*)
department(*dept_name*, *building*, *budget*)
course(*course_id*, *title*, *dept_name*, *credits*)
instructor(*ID*, *name*, *dept_name*, *salary*)
section(*course_id*, *sec_id*, *semester*, *year*, *building*, *room_number*, *time_slot_id*)
teaches(*ID*, *course_id*, *sec_id*, *semester*, *year*)
student(*ID*, *name*, *dept_name*, *tot_cred*)
takes(*ID*, *course_id*, *sec_id*, *semester*, *year*, *grade*)
advisor(*s_ID*, *i_ID*)
time_slot(*time_slot_id*, *day*, *start_time*, *end_time*)
prereq(*course_id*, *prereq_id*)

Figure 2.8 Schema of the university database.